

Motivation & Challenge

MR-only radiotherapy needs synthetic CT with accurate Hounsfield Units while preserving MRI anatomy.

The core trade-off

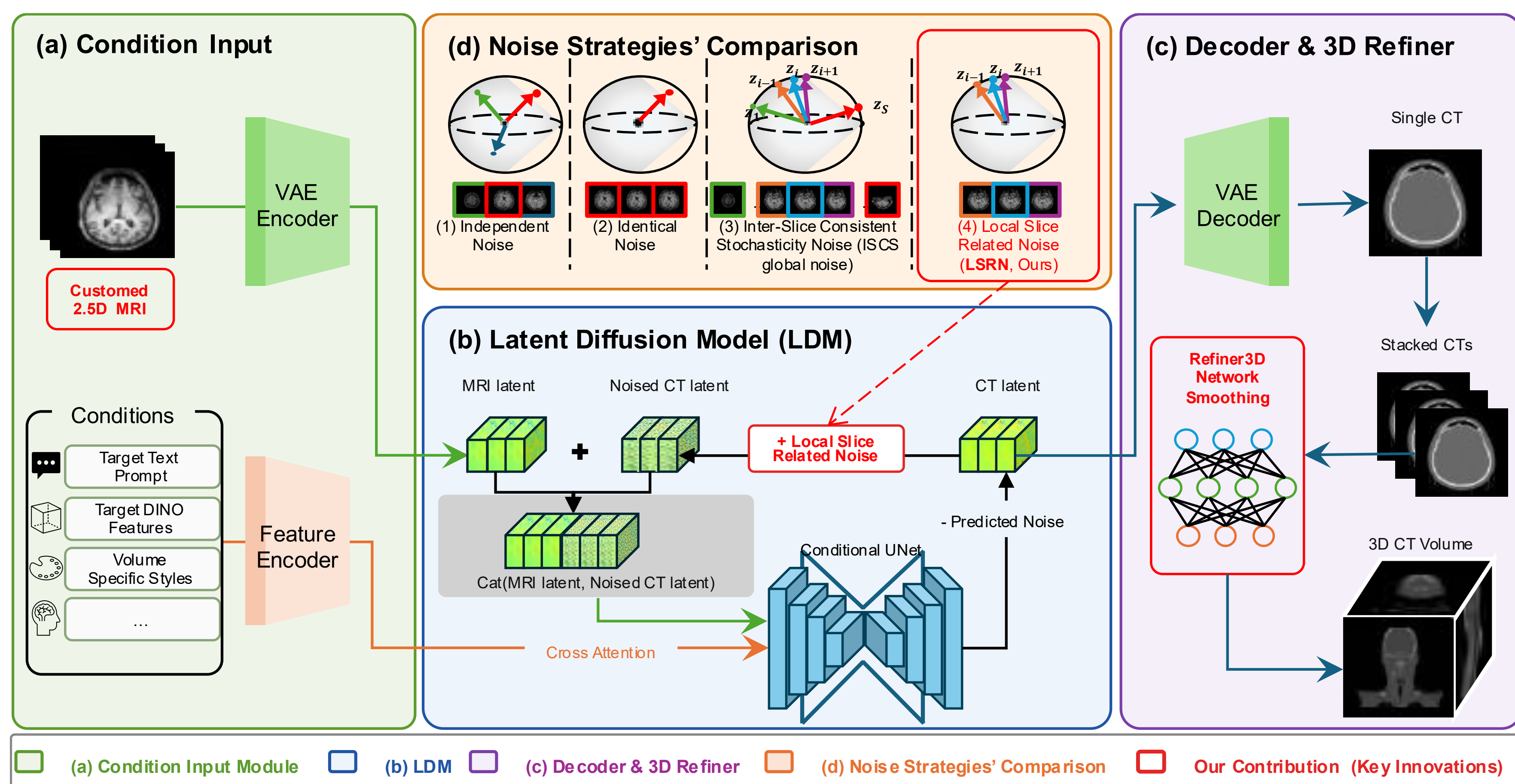
- Slice-wise 2D models are efficient, but stacking can cause inter-slice discontinuities.
- Full 3D diffusion captures volume context at high memory and compute cost.
- Existing methods mainly enforce global noise consistency, rather than local anatomical correlations.

Goal: obtain 3D-consistent MR→CT synthesis without paying the cost of full 3D diffusion.

Three key ideas

- ① **Customized 2.5D MRI conditioning**
- ② **Local Slice-Related Noise (LSRN)**
- ③ **Lightweight 3D residual refinement**

Method: Anatomy-Guided Slice-Consistent Latent Diffusion



2.5D anatomy context → Locally correlated latent noise → 3D residual refinement

Main Results

31.40 dB

PSNR
best overall

0.9018

SSIM
best overall

19.77 HU

Z-error
best overall ↓

0.0441

RMSE
best overall ↓

1,129 volumes

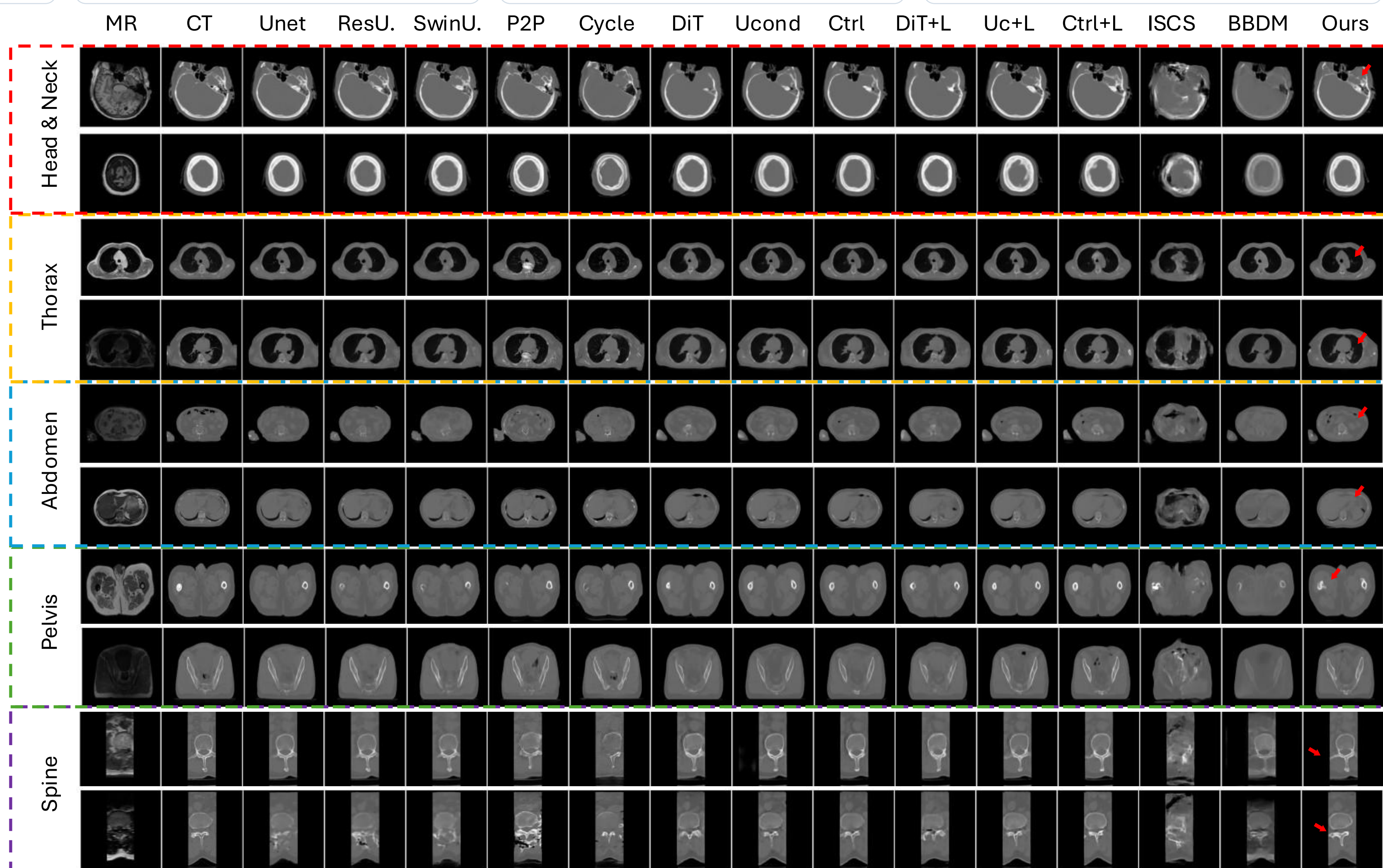
Dataset
multi-anatomy paired MRI/CT

Method	PSNR ↑	SSIM ↑	Z-error ↓
U-Net	30.33	0.8898	20.91
LDM-DiT	28.84	0.8609	23.67
ControlNet-LDM	28.97	0.8633	23.44
BBDM	21.15	0.7873	30.69
Ours	31.40	0.9018	19.77

Take-home result

Our method leads in RMSE, PSNR, SSIM and z-axis error across CNN, GAN and diffusion baselines. U-Net has slightly lower MAE (0.0147 vs 0.0150).

Key observation: local stochastic correlation improves continuity without forcing all slices to share identical noise.



Qualitative comparison across head & neck, thorax, abdomen, pelvis, and spine. Our results preserve sharper bone boundaries and more stable cross-slice anatomy.

Data & Evaluation

Data

1,129 paired MRI/CT volumes from public and private cohorts, covering head/head-neck, thorax, abdomen, pelvis and spine.

- **Paired CT and MRI Dataset:** <https://www.kaggle.com/datasets/29c3607295965ebb030f2d158fec487412d84c82528dd44f8ef956aef35541aa>
- **Gold Atlas:** <https://zenodo.org/records/583096>
- **SynthRAD2023:** <https://zenodo.org/records/7260705>
- **SynthRAD2025:** <https://zenodo.org/records/15373853>

Evaluation

- Image fidelity: MAE, RMSE, PSNR, SSIM
- Volumetric coherence: first-order z-axis error
- Qualitative axial/coronal/sagittal inspection

Training strategy

One unified model is trained progressively across all anatomies rather than separate site-specific models.

Ablation & Design Choice

PSNR (dB)

Full	31.40
w/o 2.5D	29.22
w/o LSRN	29.37
w/o Refiner	26.42

p sensitivity

Performance improves as correlation increases from $p=0$ to 0.75. At $p=1$, z-error is lowest but image fidelity drops; $p=0.75$ provides the best balance.

**Best setting: 5-slice 2.5D input
 $p = 0.75$ + 3D refiner**

Efficiency, Conclusion & Code

Efficiency vs ISCS

Input memory	1.25 MB	vs. 24.0 MB
Inference time	629 ms	vs. 883 ms
Relative input cost	1×	vs. 19.2×

Conclusion

Local anatomical conditioning + locally correlated diffusion noise + lightweight 3D refinement provide a practical middle ground between 2D efficiency and 3D coherence for multi-anatomy MR-to-CT synthesis.

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